Santanu Sarkar

Frontend Architect · Senior Frontend Engineer

Kolkata, West Bengal, India | +91 9804 243 159 | sarkarsantanu69@gmail.com
linkedin.com/in/santanusarkar86 | santanu-portfolio-frontend.vercel.app

PROFESSIONAL SUMMARY

Frontend Architect with 14+ years of experience delivering enterprise web platforms for clients in India and Singapore. Specialises in React, Next.js (App Router) and Vue 3 component systems on headless Drupal, with a focus on performance, accessibility and maintainable architecture. Also designs, builds and operates three multi-tenant SaaS products end to end, covering data model, authentication, testing, deployment and production incidents, using AI coding assistants inside a reviewed engineering workflow.

TECHNICAL SKILLS

Languages & Markup	HTML5, CSS3 / SCSS, JavaScript (ES6+), TypeScript
Frameworks	React, Next.js (App Router), Vue 3, Nuxt
Styling & Design	Tailwind CSS, Bootstrap 5, GSAP, Figma
Backend & Platform	Headless Drupal, REST / JSON:API, SEO / Core Web Vitals, AI dev tools
Full-stack (own products)	Server Actions / route handlers, Prisma, Drizzle ORM, Neon Postgres, Auth.js, Vitest, GitHub Actions, Vercel
AI & Tooling	Claude Code, GitHub Copilot, Cursor, Git / GitHub / Bitbucket, Chrome DevTools, Lighthouse

PROFESSIONAL EXPERIENCE

Frontend Developer to Frontend Architect
Mar 2015 – Present

Webspiders Pvt. Ltd · Kolkata & Singapore

- Led end-to-end frontend for enterprise and SaaS products.
- Translated Figma systems into modular React and Vue components.
- Built headless Drupal + React / Next.js frontends on REST and JSON:API.
- Raised Lighthouse scores and tightened Core Web Vitals across projects.
- Mentored junior developers on component architecture and code quality.
- Implemented SEO foundations: semantic HTML, structured metadata, Open Graph and server-side rendering.
- Worked with UX designers, backend engineers and product managers through UAT and production releases.

Founder & Product Engineer
2026 – Present

Nexvora Technologies · Kolkata

Built solo with AI coding assistants (Claude Code). I owned the product design, architecture and data model, directed the implementation, reviewed and tested the output, ran QA, and handled deployment and production incidents.

Halkhata – QR ordering and a credit ledger for local shops — Live · first real shop onboarded

halkhata.nexvoratechnologies.co.in | Next.js 15, TypeScript, Prisma, Neon Postgres, PWA, Web Push

- Server-authoritative money in integer paise, idempotent orders under a unique key, and race-safe stock decrements inside the sale transaction.
- Stateless HMAC-signed sessions verified at the Edge and re-verified against the database before any write; credential versioning revokes every device on a PIN reset.
- Wrote up a production data-loss incident and added a database-target guard that refuses production unless explicitly overridden.

EduFlow – Fees, attendance and WhatsApp reminders for coaching centres — Live · real centres

eduflow.nexvoratechnologies.co.in | Next.js 15, Drizzle ORM, Neon Postgres, Auth.js, Server Actions, PGlite tests

- Four-role multi-tenant SaaS with tenant isolation enforced on every read and write and data scoped by role on the server.
- Diagnosed and fixed a failing billing cron (Postgres 42P10 on a partial-index ON CONFLICT) that had stopped monthly fee generation for live centres.
- Integration tests run on in-memory Postgres (PGlite) built from the real SQL migrations and gate every deploy.

HealthLocker – A multi-tenant family health record platform — Pilot · demo data only

health-locker-chi.vercel.app | Next.js 16, Prisma, Neon Postgres, Column encryption, GitHub Actions CI, Capacitor

- Server-first Next.js monolith on 49 Prisma models; tenant derived from the session, AES-GCM column encryption and a nonce-based CSP.
- Found and closed two cross-tenant authorization holes and proved the fix with a role-by-guard test matrix.
- CI on every push: lint, typecheck, 250 unit tests, about 100 integration blocks, a production build and a smoke render of every page per role.

HTML Developer — Sep 2014 – Feb 2015

Budli.in

- Built mobile-optimized landing pages for paid campaigns.
- Improved product page UX, navigation and checkout flow.
- Fixed frontend and cross-browser issues across the site.

HTML Developer — Apr 2012 – Sep 2014

Delgence

- Built client websites with a focus on responsive layout.
- Applied accessibility and performance best practices.

Graphic Designer — May 2010 – Jan 2012

AD India

- Designed print and digital collateral; a grounding in typography, hierarchy and layout that carries into UI work.

SELECTED CLIENT PROJECTS

IPI Singapore ipi-singapore.org — 2024 / 25

Innovation marketplace portal — frontend architected end-to-end in Next.js App Router on headless Drupal. React · Next.js · Drupal · Tailwind · GSAP

SpiderX.ai spiderx.ai — 2023 / 24

Marketing site for a generative voice-agents platform — custom interactions, pixel-perfect delivery. HTML5 · Bootstrap 5 · jQuery · Drupal

PeerlessOne.in peerlessone.in — 2023

Financial services platform on a Vue 3 / Nuxt frontend integrated with Drupal CMS. Vue 3 · Nuxt · Bootstrap 5 · Drupal

SGInnovate sginnovate.com

Responsive frontend and custom interactions for the SGInnovate platform. HTML5 · Bootstrap 5 · jQuery · Drupal

Affinidi affinidi.com

Reusable React components integrated with a Drupal CMS backend. React · Drupal

EDUCATION & ADDITIONAL

Higher Secondary (Class XII) – Ultadanga Govt. High School, Kolkata — 2001

Secondary (Class X) – Ultadanga Govt. High School, Kolkata — 1999

Certifications: Web Design, Digital Marketing

Languages: Bengali (native), Hindi (fluent), English (professional working)